Exponential parity sample complexity below the mean-field scaling

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October 15, 2026

Abstract

Parity is easy to compute: the XOR of d input bits has a Boolean circuit with $O(d)$ gates and a polynomial-size, polynomial-weight tanh network. We show that a family of Bayesian neural-network learners nevertheless requires exponentially many examples to learn it. The networks have fixed depth, independent Gaussian weights, and a width-dependent prior specified by an exponent γ. At every fixed depth and output margin there are $c > 0$ and d_0 such that, for every $d \geq d_0$ and every fixed $\gamma < 1$, each sample size $n \leq e^{cd}$ leaves at least $1/16$ of Boolean inputs without correct-label posterior probability greater than $2/3$, with probability at least $3/4$ over uniform independent training inputs. This conclusion takes width to infinity before dimension or sample size grows. The prior is conditioned on a fixed absolute output margin at every input, and training conditions on the observed labels. We prove the bound for that posterior, including mixtures of tied minimum-energy sign functions. The proof combines the network's Gaussian width limits with kernel spectra, uniform sampling bounds, and estimates of parity correlation.

Introduction

A target function may have a small neural-network implementation without being easy to learn under a particular neural-network prior. The statistical question is whether the prior gives enough preference to simple implementations that their labels can be inferred from few examples. Parity provides a useful test. On d Boolean inputs, it is computed by a circuit with $O(d)$ gates; it also has a polynomial-size, polynomial-weight tanh implementation even at fixed depth. A learning guarantee polynomial in these complexity measures would therefore learn parity from polynomially many examples.

We prove that such a guarantee fails for the family of priors studied here. The learner is a fully connected tanh network with Gaussian weights and a linear readout. Its prior variances depend on width N through an exponent γ, with the exact scaling given in (1). We call the range $\gamma < 1$ *subcritical*. For every exponent in this range and every fixed depth, full parity requires $\exp(\Omega(d))$ examples under the width-first limiting learner. More precisely, below that sample scale, a constant fraction of test inputs fails the requirement that the correct label have posterior probability strictly greater than $2/3$.

The conditioning and prediction conventions matter. Before training, we restrict the prior to networks whose real output has absolute value greater than a fixed margin at every Boolean input. We then condition on agreement with the observed training labels. Prediction uses the posterior probability of a label. These conventions give a different learning problem from Gaussian regression on exact real-valued observations or prediction by a posterior mean. Our lower bound applies to the resulting label probabilities themselves.

The order of limits also matters. We first send width to infinity at a fixed input dimension and a fixed training sample; only afterward do we study increasing dimension and sample size. The theorem covers every specified sample size below the exponential threshold, without assuming that error is monotone in the number of examples. It concerns full parity on all d coordinates. It does not establish the conjectured $\exp(\Omega(k\log(ed/k)))$ scale for parity on k coordinates inside a larger ambient dimension d.

The proof identifies three limiting posteriors within the subcritical range. Negative exponents give a Gaussian process conditioned on the training signs; exponent zero retains the global margin condition; positive exponents concentrate on sign functions of minimum Gaussian energy. We treat these cases separately after developing their common spectral and sampling estimates. In the last case, the argument allows arbitrary posterior weights among tied minimizers, which is necessary for a conclusion about posterior confidence.

1 Model and main theorem

Let $\mathcal{X} = \{-1,1\}^d$, with uniform probability measure, and let $f(x) = \prod_{j=1}^d x_j$. We study the following precisely specified prior. All hidden layers have width N, the depth $L \geq 2$ is fixed, and

$$h^0(x) = (1,x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

There are no further biases. All weights are independent centered Gaussians, with variances

$$\operatorname{Var} W^1_{ij} = N^{1-\gamma}/d, \qquad \operatorname{Var} W^s_{ij} = N^{-\gamma} \quad (1 < s < L), \qquad \operatorname{Var} W^L_i = N^{-\gamma-1}. \tag{1}$$

Fix a margin $a > 0$; the convention $a = 1/2$ is included. Before training, condition on $|z_N(x)| > a$ for every $x \in \mathcal{X}$. Given independent uniform inputs $S = (X_1, \dots, X_n)$, subsequently condition on $\operatorname{sign} z_N(X_i) = f(X_i)$ for all i. These two conditioning operations are equivalent to conditioning on their intersection.

Define the correct-label posterior probability and its confidence-failure fraction by

$$q_{N,S}(x) = \mathbb{P}\{f(x)z_N(x) > 0 \mid |z_N(y)| > a \ \forall y;\ f(X_i)z_N(X_i) > 0 \ \forall i\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \leq 2/3\}.$$

Thus a prediction succeeds only when its actual posterior probability of being correct is strictly greater than $2/3$. This includes equality among failures. Expectations over a fresh input always use the uniform cube distribution independently of the training inputs.

Theorem 1.1. *For every fixed $L \geq 2$ and $a > 0$, there are constants $c > 0$ and d_0 such that, for every $d \geq d_0$, every integer $0 \leq n \leq e^{cd}$, and every fixed $\gamma < 1$,*

$$\mathbb{P}_S\left\{\liminf_{N\to\infty} F_N(S) \geq \frac{1}{16}\right\} \geq \frac{3}{4}.$$

The conditioning probabilities are positive for all sufficiently large N, at each fixed d, L, a, γ, S.

The width limit is taken at fixed dimension and fixed training sample. The exponential statement then concerns full parity as $d \to \infty$. No joint limit or quantitative minimum width is asserted here. The probability guarantee holds for each specified sample size with uniform constants; it does not assert a single event for every prefix of an infinite sample sequence. In particular, the proof does not assume monotonicity of this learner's fixed-target risk in sample size.

Appendix A supplies the small polynomial-weight tanh realization of parity used to compare this lower bound with implementation complexity. The obstruction is a property of the specified Bayesian learner: identifying parity from examples is not intrinsically exponentially difficult.

1.1 Proof structure and the three limiting posteriors

Write $\Psi(t) = 2\arcsin(t)/\pi$ and $\beta = 2^{-(L-1)}$. The relevant covariance is

$$K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1+x\cdot y}{d+1}\right).$$

All population function norms use the uniform probability measure on the cube. Write $\Pi_{\le r} g = g_{\le r}$ for the Walsh projection to degrees at most r, and $g_{>r} = g - g_{\le r}$. For the positive-definite covariance K, the Gaussian energy norm is $\|g\|_K^2 = g^{\mathsf{T}} K^{-1} g$; this uses the covariance matrix itself, without an extra factor 2^{-d}. At $\gamma < 0$ the limit is a Gaussian process conditioned on the training signs. At $\gamma = 0$ it retains the fixed absolute margin at every input. For $0 < \gamma < 1$ the actual network posterior concentrates on compatible globally separated functions of minimum Gaussian energy; the proof needs no formula for the weights of ties.

The proof has three tasks. First, we derive these width limits from the network prior and estimate the Fourier spectrum of K. Second, we control low-degree functions chosen after seeing the training sample. For this we prove a uniform squared negative-part estimate, using the stronger metric bound inside Lee's chaining argument [3]. A discrete Laplacian identity converts the resulting L^2 control of a degree-r polynomial into a clipped parity-correlation bound with factor r/d. A short additional band of Fourier levels covers the gaps between adjacent sampling thresholds. Third, we turn these estimates into confidence lower bounds for each of the three posteriors. The case $\gamma = 0$ requires separate Gaussian entropy bounds and an auxiliary independent sample to retain the exact global margin conditioning.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$\Psi(t) = \frac{2}{\pi}\arcsin t, \qquad K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1+x\cdot y}{d+1}\right). \tag{2}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x)\ge a\ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{3}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1+\epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N\to\infty} N^{-\gamma} \log \mathbb{P}\{g(x) z_N(x) > a\ \forall x\} = -E_K(g). \tag{4}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1}\sum_i h_i^s(x) h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \le C_\epsilon e^{-c_\epsilon N} \tag{5}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma}Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1 + x \cdot y)/d$. Because $N^{1-\gamma} \to \infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[\Psi\left(\frac{Q(x,y)}{\sqrt{Q(x,x)Q(y,y)}}\right)\right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there. Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (5). The first-layer limit normalizes the diagonal $(d + 1)/d$, giving the kernel in (2).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma}Q_N^{L-1}). \tag{6}$$

Couple it to $Y_N = N^{-\gamma/2}K^{1/2}\xi$, where $\xi \sim \mathcal{N}(0, I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (4), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

Proposition 2.2 (The nonpositive-exponent posterior limits)**.** *For $\gamma = 0$, the Boolean posterior converges to that of $G \sim \mathcal{N}(0, K)$ conditioned on $|G(x)| > a$ everywhere and on the observed signs. For $\gamma < 0$, it converges to the Gaussian posterior conditioned only on the observed signs.*

Proof. The hidden-kernel argument establishing (5) holds for every $\gamma < 1$. By (6), the normalized vector $N^{\gamma/2}Z_N$ therefore converges in law to $\mathcal{N}(0, K)$. Its all-input margin is $aN^{\gamma/2}$, which equals a when $\gamma = 0$ and tends to zero when $\gamma < 0$. The limiting Gaussian is nonsingular. Thus all boundaries involved in any fixed full-sign event or margin event have probability zero, and the conditioning events have positive probability. Taking ratios of their probabilities gives convergence of each Boolean posterior mass. There are finitely many masses for fixed d. □

3 Fourier level masses of the kernel

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and $F = \Psi^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j\in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work

on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this section and in Section 5 may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 3.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \geq 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of Ψ has positive coefficients at every odd exponent, with

$$[z^{2k+1}]\Psi(z) = \frac{2}{\pi} \frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$\Psi(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^\beta + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [2, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|\Psi(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = {} & j[p_m(j-1) - p_m(j+1)] \\ & + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1}+\binom{d}{j}=\binom{D}{j}$. Consequently

$$D_r=\sum_{\substack{j\le r\\ j \text{ odd}}}\binom{D}{j},\quad \tau_r=\sum_{\substack{j>r\\ j \text{ odd}}}\binom{D}{j}a_j,\quad \lambda_{r+1}=a_{r+2}. \tag{7}$$

The odd cutoff is essential for these exact identities.

Lemma 3.2 (Sharp level and tail estimates)**.** *Uniformly for odd $1\le j\le D/4$ and odd $1\le r\le D/8$,*

$$\binom{D}{j}a_j \asymp j^{-1-\beta}, \tag{8}$$

$$\tau_r \asymp r^{-\beta}, \tag{9}$$

$$\frac{\lambda_{r+1}}{\tau_r}\asymp \frac{1}{r\binom{D}{r+2}}, \tag{10}$$

$$s_{2,r}:=\sum_{q>r}\binom{d}{q}\lambda_q^2\le \frac{Cr^{-2\beta-2}}{\binom{D}{r+2}}. \tag{11}$$

The same level-mass estimate holds up to $j\le \rho D$ for any fixed $\rho<1/2$, with constants depending on ρ.

Proof. The gamma integral and Lemma 3.1 give, for odd j,

$$a_j \asymp D^{-\beta}\int_0^\infty u^{-1-\beta}p(u)^j(1-p(u))^{D-j}\,du,\quad p(u)=\frac{1-e^{-2u}}{2}, \tag{12}$$

$$\asymp D^{-\beta}\int_0^{1/2} p^j(1-p)^{D-j}w_\beta(p)\,dp, \tag{13}$$

where $w_\beta(p)=[-\frac12\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0,b]$ for every fixed $b<1/2$.

Fix $\rho<b<1/2$ and take $j\le\rho D$. The integral on $[0,b]$ is comparable to the complete beta integral $B(j-\beta,D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1-\rho/b$ of its mass in $[0,b]$. The identity

$$\binom{D}{j}B(j-\beta,D-j+1)=\frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)}\asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b,1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (8) and its stated extended range.

Sum (12) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r\le CD^{-\beta}\int_0^\infty u^{-1-\beta}\min\{1,Du/(r+1)\}\,du\le C(r+1)^{-\beta}.$$

For the lower bound, sum (8) over the odd j in $(r,2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (10) follows from (8), (9), and (7).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (11). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{14}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

Corollary 3.3 (Lower bounds used by the interpolation)**.** *For sufficiently large d,*

$$\lambda_0 \ge c/d, \qquad \lambda_j \ge \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (0 \le j \le .49d), \qquad v := \lambda_{\min}(K) \ge c(d\log d)^{-\beta}.$$

Also $v \to 0$.

Proof. The first bound follows from the level estimate at full-cube degree one. For odd reduced degree j, use $\lambda_j = a_j$ and $\binom{D}{j}/\binom{d}{j} = D/(D-j)$. For even j, use $\lambda_j = a_{j+1}$ and $\binom{D}{j+1}/\binom{d}{j} = D/(j+1)$. The extended level bound with any fixed cutoff between .49 and 1/2 then gives the stated uniform lower bound, including $j = 0$.

For any odd full-cube degree $j \le D$, restrict the integral (12) to $u \ge \frac{1}{2}\log(2D)$. In this range $p(u) \ge (1 - 1/(2D))/2$ and $1 - p(u) \ge 1/2$, so $p(u)^j(1-p(u))^{D-j} \ge c2^{-D}$. Integrating $u^{-1-\beta}$ gives $a_j \ge c2^{-D}D^{-\beta}(\log D)^{-\beta}$. Every reduced eigenvalue equals an odd a_j, and the matrix eigenvalues are M times the normalized ones. Thus $v \ge c(d\log d)^{-\beta}$. Finally, for any fixed small positive cutoff fraction, the minimum normalized eigenvalue is at most the average in its tail, whence $v \le M\tau_r/(M - D_r) = O(d^{-\beta})$. This proves $v \to 0$. □

4 One-sided squared sampling with one logarithm

The sampling estimate needed below concerns a squared one-sided loss. It follows from a metric estimate in the proof of Lee's chaining theorem [3, Lemma 2.12, equation (2.20)]. We spell out the comparison: a first-power ReLU sampling theorem alone would not give this result. Throughout this section, C denotes a universal constant that may change from line to line, and $t_- := \max\{-t, 0\}$.

Lemma 4.1 (A squared Lipschitz process)**.** *Let $a_1, \dots, a_m \in \mathbb{R}^D$, and set*

$$R = \max_{i \le m} \|a_i\|_2, \qquad V = \sup_{\|u\|_2 \le 1} \left(\sum_{i=1}^m \langle a_i, u\rangle^2 \right)^{1/2}.$$

For each i, let $\psi_i : \mathbb{R} \to \mathbb{R}$ satisfy $\psi_i(0) = 0$ and $\mathrm{Lip}(\psi_i) \le 1$. For independent uniform signs ϵ_i,

$$\mathbb{E}_\epsilon \sup_{\|u\|_2 \le 1} \left| \sum_{i=1}^m \epsilon_i \psi_i(\langle a_i, u\rangle)^2 \right| \le CR\sqrt{\log(2(m+D))}\, V. \tag{15}$$

Proof. Use the genuine seminorms $N_i(u) = |\langle a_i, u\rangle|$ and the symmetric quasimetric

$$d_A(u,v)^2 = \sum_{i=1}^{m} N_i(u-v)^2 (N_i(u)^2 + N_i(v)^2).$$

The stronger intermediate estimate established in the proof of [3, Lemma 2.12] is

$$\gamma_2(T, d_A) \le C(\lambda\sqrt{\log(2D)} + \kappa) \sup_{u\in T} \left(\sum_i N_i(u)^2\right)^{1/2}, \qquad T \subseteq B_2^D, \tag{16}$$

where

$$\lambda = \max_i (\mathbb{E} N_i(g)^2)^{1/2}, \qquad \kappa = \mathbb{E} \max_i N_i(g), \qquad g \sim \mathcal{N}(0, I_D).$$

Here γ_2 is the generic-chaining functional. This is equation (2.20) of that proof, with the energy normalization (2.17) undone. Seminorms follow from the norm statement by replacing $N_i(u)$ with $(N_i(u)^2 + \varepsilon^2\|u\|_2^2)^{1/2}$ and then letting $\varepsilon \downarrow 0$; the resulting metrics dominate d_A, and the parameters on the right converge. For our rank-one seminorms, $\lambda = R$ and $\kappa \le R\sqrt{2\log(2m)}$.

Let $X_u = \sum_i \epsilon_i \psi_i(\langle a_i, u\rangle)^2$. Its subgaussian increment metric is bounded by $\sqrt{2}\, d_A$, because

$$\begin{aligned}
&\sum_i \left(\psi_i(\langle a_i, u\rangle)^2 - \psi_i(\langle a_i, v\rangle)^2\right)^2 \\
&\quad \le \sum_i |\langle a_i, u-v\rangle|^2 \left(|\langle a_i, u\rangle| + |\langle a_i, v\rangle|\right)^2 \le 2 d_A(u,v)^2.
\end{aligned}$$

We used both Lipschitz continuity and $|\psi_i(t)| \le |t|$. The subgaussian chaining inequality, together with $X_0 = 0$, bounds $\mathbb{E}\sup X_u$ by $C\gamma_2(B_2^D, d_A)$. Apply the same argument to $-X_u$ and add. Equation (16) now proves (15). □

Remark 4.2 (The normalization in the metric estimate). The proof of [3, Lemma 2.12] passes through a coarse additive constant in its equation (2.26), after normalizing $\sup_T \sum_i N_i^2 = 1$. The homogeneous version (16) follows by retaining the scale in the two geometric terms of that argument. Indeed, in the rank-one case $T \subseteq B_2^D$ implies $N_i(u) \le \lambda$ on T; writing $\|u\|_{\mathcal{N}} = \max_i N_i(u)$ gives $d_A(u,v) \le \sqrt{2}\|u - v\|_{\mathcal{N}}$ on T. Thus $\operatorname{diam}_{d_A}(T) \le C\lambda$ and the volumetric packing constant in its Corollary 2.11 can be taken to be $C\lambda$, since $T \subseteq \lambda B_{\mathcal{N}}$. The additive term is therefore $C\lambda$, absorbed by $C(\lambda\sqrt{\log(2D)} + \kappa)$. The growth-functional portion of the proof is unchanged. If the packing constant is less than one, first rescale the quasimetric to apply that corollary in its stated normalization, and then scale back. Most importantly, throughout that argument the N_i are symmetric seminorms. No assertion that a positive part is a seminorm is used.

Lemma 4.3 (Uniform one-sided squared sampling)**.** *Let (Z, Y) be a random pair with $Z \in \mathbb{R}^D$, $Y \in \{-1, 1\}$, $\mathbb{E} ZZ^{\mathsf{T}} = I_D$, and $\|Z\|_2^2 = D$ almost surely. For independent copies (Z_i, Y_i), let*

$$\Delta_n = \sup_{\|u\|_2 \le 1} \left| \frac{1}{n} \sum_{i=1}^{n} (Y_i \langle Z_i, u\rangle)_-^2 - \mathbb{E}(Y\langle Z, u\rangle)_-^2 \right|, \qquad h_n = \sqrt{\frac{D\log(2(n+D))}{n}}.$$

Then $\mathbb{E}\Delta_n \le C h_n(1 + C h_n)$. In particular, for any fixed $\varepsilon, \delta > 0$ there is a finite $A_{\varepsilon,\delta}$ such that

$$n \ge A_{\varepsilon,\delta} D \log(2D) \quad \Longrightarrow \quad \mathbb{P}\{\Delta_n > \varepsilon\} \le \delta.$$

The labels may depend arbitrarily on the features.

Proof. Put $Q_n = n^{-1}\sum_i Z_i Z_i^{\mathsf{T}}$. Conditional on the observations, apply Lemma 4.1 to $a_i = Y_i Z_i/\sqrt{n}$ and $\psi_i(t) = t_-$. Here $R = \sqrt{D/n}$ and $V = \sqrt{\|Q_n\|_{\mathrm{op}}}$. Symmetrization gives

$$\mathbb{E}\Delta_n \le Ch_n\,\mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}}. \tag{17}$$

We also apply the same process lemma with $\psi_i(t) = t$. Since $\mathbb{E}Q_n = I_D$, symmetrization and Jensen's inequality imply, with $A = \mathbb{E}\|Q_n\|_{\mathrm{op}}$,

$$A \le 1 + \mathbb{E}\|Q_n - I_D\|_{\mathrm{op}} \le 1 + Ch_n\mathbb{E}\sqrt{\|Q_n\|_{\mathrm{op}}} \le 1 + Ch_n\sqrt{A}.$$

Solving this quadratic inequality yields $\sqrt{A} \le 1 + Ch_n$, after increasing C. Substitute into (17). Finally, at $n \ge A_{\varepsilon,\delta} D\log(2D)$ the quantity h_n can be made uniformly small by increasing $A_{\varepsilon,\delta}$. Markov's inequality proves the probability statement. It is a statement for each specified sample size n. □

Lemma 4.4 (Parity balances squared amplitudes)**.** *Let $f(x) = \prod_{j=1}^d x_j$ on the uniform discrete cube, and let P have Walsh degree at most r. Then*

$$\mathbb{E}(fP)_-^2 \ge \left(\frac{1}{2} - \sqrt{\frac{r}{d}}\right)\|P\|_2^2. \tag{18}$$

Consequently, for each fixed $\delta > 0$ there is a universal A_δ such that, if $r \le d/64$ and $n \ge A_\delta D_r \log(2D_r)$, where $D_r = \sum_{j=0}^r \binom{d}{j}$, then, with probability at least $1-\delta$,

$$\frac{1}{n}\sum_{i=1}^n (f(X_i)P(X_i))_-^2 \ge \frac{1}{4}\|P\|_2^2 \quad \textit{for every polynomial } P \textit{ of degree at most } r. \tag{19}$$

Proof. Write $x^{(j)}$ for x with coordinate j flipped, and $D_jP = (P(x) - P(x^{(j)}))/2$. Parity changes sign under each flip. For $F(t) = t|t|$, the inequality $|F(a) - F(b)| \le (|a| + |b|)|a - b|$ and Cauchy–Schwarz give

$$|\mathbb{E}[fP|P|]| = \frac{1}{2}\left|\mathbb{E}f(x)(F(P(x)) - F(P(x^{(j)})))\right| \le 2\|P\|_2\|D_jP\|_2.$$

Average over j and use $\sum_j \|D_jP\|_2^2 \le r\|P\|_2^2$ to obtain $|\mathbb{E}[fP|P|]| \le 2\sqrt{r/d}\,\|P\|_2^2$. Since $(fP)_-^2 = (P^2 - fP|P|)/2$, this proves (18).

The feature vector consisting of all Walsh characters of degree at most r is isotropic and has squared norm exactly D_r. When $r \le d/64$, the population lower bound is at least $3\|P\|_2^2/8$. Apply Lemma 4.3 with $Y = f(X)$ and $\varepsilon = 1/8$, and use homogeneity, to obtain (19). □

5 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 5.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \eta_r := C\sqrt{D}\,e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h; D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b(1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h; D)|}{\binom{D}{j}} \le C\sqrt{D}\exp[-2\delta j(1-j/D)]. \tag{20}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h; D)$. For $r < j \le D - r$, (20) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2}\exp[-D(\mathsf{h}(1/4) - \mathsf{h}(1/8)) + O(1)].$$

Here $\mathsf{h}(t) = -t\log t - (1-t)\log(1-t)$ denotes the binary entropy in nats. The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 5.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\text{near}} = \#\{y : \text{dist}(x,y) \notin [d/4, 3d/4]\} \le 2e^{d\mathsf{h}(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned}&\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\text{op}} > z\tau\}\\ &\quad\le \frac{C}{z^2}\left[\frac{q\eta_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\text{near}})}{M}\right]^2 + \frac{n^2(1+B_{\text{near}})}{M}.\end{aligned} \tag{21}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned}\|J_{\text{far}}\|_{\max} &\le \eta_r,\\ \|J_{\text{far}}\|_{1\to 2}^2 &\le Ms_2,\\ \|J_{\text{far}}\|_{\text{op}} &\le M\ell + \tau + \tau B_{\text{near}}.\end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [4, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\eta_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\mathrm{near}})]. \tag{22}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (22) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (21). □

6 Clipped parity and adaptive spectral cutoffs

All function norms in this section use the uniform cube probability measure. Write $\mathrm{clip}(t) = \max\{-1, \min\{1, t\}\}$ and $\Delta = \sum_{j=1}^{d}(I - \mathrm{flip}_j)/2$. Thus Δ has eigenvalue j on Fourier level j.

Lemma 6.1 (Clipped parity correlation)**.** *For any real cube function P and any $b > 0$,*

$$|\mathbb{E} f\,\mathrm{clip}(P/b)| \le \frac{\|\Delta P\|_1}{bd}. \tag{23}$$

In particular, if P has degree at most u, the right side is at most $u\|P\|_2/(bd)$. If P has degree at most u, $z = P + t$, and $|z| \ge b$ everywhere, then

$$|\mathbb{E} f\,\mathrm{sign}\, z| \le \frac{u\|P\|_2}{bd} + \frac{\|t\|_2}{b}. \tag{24}$$

Proof. Let R flip one uniformly chosen coordinate, put $q = fP$, and set $e = 2f\Delta P/d$. Since f changes sign along every edge, $Rq = -q + e$. Put $A = \mathbb{E}|q + b|$ and $B = \mathbb{E}|q - b|$. Jensen's inequality and invariance of the uniform measure give $\mathbb{E}|Rq + b| \le A$, whereas the triangle inequality gives $\mathbb{E}|Rq + b| \ge B - \|e\|_1$. Replacing b by $-b$ gives the opposite inequality, so $|A - B| \le \|e\|_1$. The identity

$$\mathrm{clip}(q/b) = \frac{|q + b| - |q - b|}{2b}$$

proves (23). For a degree-u function, $\|\Delta P\|_1 \le \|\Delta P\|_2 \le u\|P\|_2$. Finally, $\mathrm{sign}\, z = \mathrm{clip}(z/b)$ under the margin assumption. Clipping is 1-Lipschitz, so its difference from $\mathrm{clip}(P/b)$ has mean absolute value at most $\|t\|_1/b \le \|t\|_2/b$. □

The following elementary polynomial-threshold obstruction is due to Aspnes, Beigel, Furst, and Rudich [1]; we include its proof.

Lemma 6.2 (Polynomial thresholds cannot approximate parity)**.** *For $0 \le s < d$, every degree-s polynomial threshold, with a fixed common convention for zero ties, has parity error at least*

$$2^{-d} \sum_{j=0}^{\lfloor (d-s-1)/2 \rfloor} \binom{d}{j}. \tag{25}$$

This lower bound tends to $1/2$ whenever $s = o(\sqrt{d})$.

Proof. A sufficiently small constant perturbation resolves any zero ties without changing the specified threshold classifier. Thus suppose P has no zeros. Put $t = \lfloor (d-s-1)/2 \rfloor$. If its error set had fewer than $\sum_{j \le t} \binom{d}{j}$ points, a nonzero polynomial Q of degree at most t could be chosen to vanish on every error. The function fPQ^2 would then be nonnegative everywhere and positive somewhere. But PQ^2, after multilinear reduction, has degree at most $s + 2t < d$, so its parity Fourier coefficient $\mathbb{E} fPQ^2$ is zero, a contradiction. This proves (25). The last assertion follows from the central binomial bound $\max_j 2^{-d}\binom{d}{j} = O(d^{-1/2})$. □

Lemma 6.3 (The small-sample cutoff)**.** *Let s be an odd integer with $s \sim d^{1/3}$. For every fixed $C, m > 0$, uniformly over $0 \le n \le \exp[(\log(d+1))^3]$,*

$$C d^m (1+n) \lambda_{s+1} = o(1).$$

Consequently, a globally margin-a function z satisfying $\|z\|_K^2 \le C d^m (1+n)$ has parity error at least $1/2 - o(1)$, uniformly over such functions and sample sizes. Here $\|z\|_K^2 = z^{\mathsf{T}} K^{-1} z$.

Proof. Lemma 3.2 gives $\lambda_{s+1} \le C s^{-1-\beta} / \binom{d+1}{s+2}$. The logarithm of this binomial coefficient is of order $d^{1/3} \log d$, which dominates $(\log(d+1))^3 + O(\log d)$. For $P = z_{\le s}$, spectral orthogonality therefore gives $\|z - P\|_2^2 \le \lambda_{s+1} \|z\|_K^2 = o(1)$. Because $|z| \ge a$, the signs of z and P can disagree on at most $\|z - P\|_2^2 / a^2 = o(1)$ of the cube. Apply Lemma 6.2 with $s = o(\sqrt{d})$. □

For an odd integer r, retain the notation $D_r, \tau_r, s_{2,r}$ from (7) and Lemma 3.2, and write $\ell_r = \lambda_{r+1}$.

Lemma 6.4 (Adaptive cutoff and a short additional band)**.** *Fix $A \ge 1$, $\delta > 0$, and $0 < \alpha < 1/256$. There exists $c > 0$ such that the following holds uniformly for*

$$\exp[(\log(d+1))^3] \le n \le e^{cd}$$

as $d \to \infty$. Let r be the largest odd integer satisfying $A D_r \log(2D_r) \le n$, and put

$$\rho = \frac{r}{d+1}, \qquad H_r = \log(e/\rho), \qquad u = r + 2\lceil 10 \log(d+1) \rceil.$$

Then

$$\tfrac{1}{4}(\log d)^2 \le r \le \alpha d, \quad u/r = 1 + o(1), \quad u \le d/64, \tag{26}$$

$$n < A D_{r+2} \log(2D_{r+2}), \qquad \frac{n \ell_r}{\tau_r} \le C_L A H_r, \tag{27}$$

$$\frac{n d\, s_{2,r}}{\tau_r^2} \le C_L A \rho^{-1} H_r, \qquad \frac{\tau_u}{\tau_r} = 1 + o(1), \qquad n \ell_u \le d^{-10} \tau_r. \tag{28}$$

Moreover, with probability at least $1 - \delta$,

$$\|(K_{>r})_{SS}\|_{\mathrm{op}} \le b_r \tau_r, \qquad b_r = C_{L,\delta} \left[1 + \sqrt{A \rho^{-1} H_r} + A H_r \right]. \tag{29}$$

The estimates and the probability are uniform over each specified n in the range; no simultaneous event over all n is asserted.

Proof. The bound $D_r \le (ed/r)^r$ for $r \le d/2$, with an immaterial constant enlargement at bounded r, shows that the sampling threshold is met by an odd integer just above $(\log d)^2/4$. Conversely, binomial entropy bounds show that $D_r > e^{cd}$ whenever $r > \alpha d$, if c is sufficiently small. Maximality of r then gives the next-threshold inequality in (27). Using (14) and $\log(2D_{r+2}) \le Cr\log(e(d+1)/r)$ in that inequality yields

$$n \le CA\binom{d+1}{r+2} rH_r.$$

Equations (10) and (11) now prove the bounds involving ℓ_r and $s_{2,r}$.

For the additional band, apply the two-sided level estimate only at its endpoints, to obtain

$$\frac{\ell_u}{\ell_r} \le C_L \left(\frac{r+2}{u+2}\right)^{1+\beta} \frac{\binom{d+1}{r+2}}{\binom{d+1}{u+2}} \le C_L 4^{-(u-r)/2}.$$

The last inequality follows by multiplying the exact two-step binomial ratios below $(d+1)/8$. In particular the fixed comparison constant C_L is used once, not once per level. Together with (27) and the choice of u, this implies $n\ell_u \le d^{-10}\tau_r$ for all sufficiently large d. Also $0 \le \tau_r - \tau_u \le C_L(u-r)r^{-1-\beta}$, so (9) gives $\tau_u/\tau_r = 1 + o(1)$ uniformly.

Finally apply Lemma 5.2. Its far-coherence term $q\eta_r/\tau_r$ tends to zero uniformly by Lemma 5.1 and the lower bound on r. Choose c also to satisfy $2c < \log 2 - h(1/4)$, so that its near-pair and collision terms vanish. The two remaining terms are bounded by $C_L[\sqrt{A\rho^{-1}H_r} + AH_r]$. Increasing $C_{L,\delta}$ in (29) makes the failure probability at most δ for all sufficiently large d. □

7 Conditioning cost and amplitude at the Gaussian endpoint

Fix $a > 0$. For an input sample S and arbitrary labels $y_i \in \{-1, 1\}$, define the full event

$$A_S = \{|G(x)| > a \ \forall x \in \mathcal{X};\ y_i G(X_i) > 0 \ \forall i\}, \qquad \mathcal{D}(S) = -\log \mathbb{P}\{G \in A_S\}, \quad G \sim N(0, K).$$

All probabilities below concern this exact event. Write $M = 2^d$ and $v = \lambda_{\min}(K) > 0$ for the smallest *matrix* eigenvalue; it differs from the normalized Walsh eigenvalues λ_j.

Lemma 7.1 (Adaptive conditioning cost)**.** *For $1 \le n \le e^{d/25}$, with probability at least $1 - e^{-cd}$ over independent uniform inputs, simultaneously for all labelings and every odd $1 \le r \le d/200$,*

$$\mathcal{D}(S) \le C\{d^2 + nd^{1+\beta}\tau_r + D_r \log(r+1)\}, \tag{30}$$

$$\mathcal{D}(S) \le C\{d^2 + nd^{1+\beta}\}. \tag{31}$$

The constants depend only on L, a.

For a covariance J on the cube, use the Gaussian Cameron–Martin norm $\|h\|_J^2 = h^T J^{-1} h$. For a translation-invariant kernel it equals $\sum_U \widehat{h}(U)^2/\lambda^J_{|U|}$, using normalized Walsh coefficients.

To prove the lemma, let Q retain degrees greater than r, let $K_{\text{low}} = K - Q$, and set $\tau = \tau_r$. Define

$$\widetilde{K} = \tau K_{\text{low}} + Q.$$

Lemma 3.2 proves $\tau \asymp r^{-\beta}$, while Corollary 3.3 gives $\lambda_0 \ge c/d$ and $v \ge c(d\log d)^{-\beta}$. In particular $0 < \tau < 1$ for large d. The diagonal of $\widetilde{K}$ is $\tau(1-\tau) + \tau \le 2\tau$. Its low-degree eigenvalues are those

of K multiplied by τ, while all remaining eigenvalues agree. The Gaussian densities thus satisfy, pointwise,

$$\frac{dN(0,K)}{dN(0,\widetilde{K})}(g) = \tau^{D_r/2}\exp\left[\frac{1}{2}g^T(\widetilde{K}^{-1}-K^{-1})g\right] \geq \tau^{D_r/2}. \tag{32}$$

The comparison covariance is only a proof device; it does not replace the prior in the theorem.

Lemma 7.2 (Localized interpolation)**.** *For all sufficiently large d, with probability at least $1-e^{-cd}$ over an independent uniform sample with $1 \leq n \leq e^{d/25}$, simultaneously for every labeling, every odd $1 \leq r \leq d/200$, and every $A > 0$, there is a function h such that*

$$|h(x)| > A \quad \forall x, \qquad y_i h(X_i) > A \quad \forall i, \qquad \|h\|_{\widetilde{K}}^2 \leq CA^2(d/\tau + nd^{\beta}).$$

Proof. For $z \in \mathcal{X}$ define

$$b_z(x) = 20\int_{2/5}^{9/20}(1-2p)^{\mathrm{dist}(x,z)}\,dp.$$

Then $b_z(z) = 1$, $0 \leq b_z(x) \leq (1/5)^{\mathrm{dist}(x,z)}$, and its Walsh coefficients have magnitudes

$$\widehat{b}_j = 20\int_{2/5}^{9/20}(1-p)^{d-j}p^j\,dp, \qquad \binom{d}{j}\widehat{b}_j \leq \frac{20}{d+1}, \qquad \widehat{b}_j \leq (3/5)^d.$$

The middle inequality follows by extending to the complete beta integral. Corollary 3.3 gives

$$\lambda_j \geq \frac{cd^{-1-\beta}}{\binom{d}{j}} \quad (j \leq .49d).$$

Therefore the contribution of these degrees to $\|b_z\|_K^2$ is at most

$$Cd^{1+\beta}\sum_{j\leq .49d}\left[\binom{d}{j}\widehat{b}_j\right]^2 \leq Cd^{\beta}.$$

For $j > .49d$ the integrand increases with p on the integration interval, giving $\widehat{b}_j \leq .55^{d-j}.45^j$. Since every $\lambda_j \geq v/M$, the remaining energy is at most

$$\frac{2^d}{v}\sum_{j>.49d}\binom{d}{j}.55^{2(d-j)}.45^{2j} \leq \mathrm{poly}(d)e^{-c_* d},$$

where

$$c_* = -[\log 2 + h(.49) + 1.02\log(.55) + .98\log(.45)] > .0062.$$

Here $h(t) = -t\log t - (1-t)\log(1-t)$, and the binomial entropy bound applies because the exponent decreases for $j/d \geq .49$.

The low-degree energy increases by at most

$$\tau^{-1}D_r(.72)^d/v$$

under damping. Uniformly in the stated range, $D_r \leq \exp[(h(1/200)+o(1))d]$. Since $h(1/200) < .04$, this term decays exponentially. Hence $A_d := \|b_z\|_{\widetilde{K}}^2 \leq Cd^{\beta}$.

Let $J(z,w) = \langle b_z, b_w \rangle_{\widetilde{K}}$. Its normalized Fourier eigenvalues are $\nu_j = \widehat{b}_j^2/\lambda_j^{\widetilde{K}} \le .72^d/(\tau v)$. Parseval and Cauchy–Schwarz give

$$\mathbb{E}_{X,Y}|J(X,Y)| \le \left(\sum_j \binom{d}{j} \nu_j^2\right)^{1/2} \le \sqrt{\nu_{\max} A_d}.$$

For every subset T of the sampled indices, even one chosen after seeing the sample,

$$\left\| \sum_{i \in T} b_{X_i} \right\|_{\widetilde{K}}^2 \le n A_d + \sum_{i \ne j} |J(X_i, X_j)|.$$

The expected cross sum is at most $n(n-1)\sqrt{\nu_{\max} A_d}$. Markov's inequality at threshold nd^β has exponentially small failure when $n \le e^{.04d}$, because $\frac{1}{2}\log(1/.72) > .16$. Thus all subsets simultaneously have energy at most Cnd^β. A union bound over the at most d cutoffs preserves exponentially small failure; the estimates are uniform because $\tau_r \ge cd^{-\beta}$.

Also, with exponentially high probability, every pair of different sample indices has inputs at distance greater than $.2d$: a union bound costs at most $n^2 \exp[-d(\log 2 - h(.2))]$, with $\log 2 - h(.2) > .19$. This event excludes repetitions. At any input at most one center is within distance $.1d$, and the contribution of all other centers is at most $n(.2)^{.1d} = o(1)$. Thus the bump sum over any selected centers is $1 + o(1)$ at a selected center and at most $.2 + o(1)$ everywhere else.

For $T = \{i : y_i = -1\}$ take $h = 2A(1 - 2\sum_{i \in T} b_{X_i})$. The preceding bounds give its strict global and training margins. Since $\|1\|_{\widetilde{K}}^2 = 1/(\tau\lambda_0) \le Cd/\tau$, the squared triangle inequality gives the asserted energy bound. □

Corollary 7.3 (A strict-margin comparator)**.** *On the same sample event, for every labeling and every $a_0 > 0$ there is a function h with $|h(x)| > a_0$ at every input, $y_i h(X_i) > a_0$ at every training location, and*

$$\|h\|_K^2 \le C a_0^2 (d + nd^\beta).$$

Proof. Apply Lemma 7.2 at the fixed cutoff $r = 1$, for which $\tau_1 \asymp 1$. Since $\widetilde{K} \preceq K$, its Gaussian energy bounds the original energy from above: $\|h\|_K^2 \le \|h\|_{\widetilde{K}}^2$. The same event therefore also supplies the undamped comparator used below. □

Proof of Lemma 7.1. Put $t_0 = \sqrt{4\tau \log(4M)}$ and $A = a + t_0 + 1$. Under $N(0, \widetilde{K})$ the symmetric event $B = \{\max_x |G(x)| \le t_0\}$ has probability at least $1/2$, by the diagonal bound and a Gaussian union bound. Lemma 7.2 makes $h + B$ a subset of A_S. Because a is fixed, $\tau \ge cd^{-\beta}$, and $\beta < 1$, we have $A^2 \le C_a \tau d$ for large d. Hence

$$\|h\|_{\widetilde{K}}^2 \le C(d^2 + nd^{1+\beta}\tau).$$

For a symmetric event B, the Gaussian translation formula, symmetry, and $\cosh t \ge 1$ imply

$$\mathbb{P}_{\widetilde{K}}(h + B) \ge \mathbb{P}_{\widetilde{K}}(B) e^{-\|h\|_{\widetilde{K}}^2/2}.$$

Combine this with (32) and $\log(1/\tau) \le C \log(r+1)$ to obtain (30). The identical argument without damping uses K itself, a buffer of squared size $O(d)$, constant-function energy $O(d)$, and bump energy $O(nd^\beta)$, proving (31). □

7.1 Large magnitudes for actual output draws

Write Φ for the standard normal distribution function and $\overline{\Phi} = 1 - \Phi$.

Lemma 7.4 (A finite Gaussian amplitude estimate)**.** *Let G be a nondegenerate centered Gaussian vector on a finite set. Suppose an event A implies $|G(x)| > a$ at every coordinate and $-\log \mathbb{P}(A) \le D$, with $D \ge 1$. Let $T = (x_1, \dots, x_m)$ be distinct coordinates with $K_T \succeq wI_m$, where $w > 0$. Assume $D < m$ and $e^{-2a^2/w} \le 1/2$. Set*

$$t = \sqrt{\log(m/D)}, \qquad b_0 = a + \sqrt{w}\, t, \qquad c_0 = \frac{e^{-3/2}}{\sqrt{2\pi}}.$$

Then

$$\mathbb{E}\left[\frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{|G(x_i)| \le b_0/2\}} \,\middle|\, A\right] \le \frac{2}{c_0}(t+1)\frac{D+1}{\sqrt{mD}} + \frac{8w}{b_0^2} + \frac{16wD}{mb_0^2}. \tag{33}$$

The event may impose arbitrary further constraints outside T.

Proof. On an enlarged Gaussian space draw $H_T \sim N(0, K_T - wI)$ and $\xi \sim N(0, I_m)$ independently, set $G_T = H_T + \sqrt{w}\xi$, and draw the other coordinates from their original conditional law given G_T. This reproduces the original full law of G. The outside coordinates may depend on ξ; no independence from them is asserted. The event inclusion alone gives

$$\mathbb{P}(A \mid H_T) \le \prod_{i=1}^{m} B_{a,w}(H_T(i)), \quad B_{a,w}(h) = \Phi((h-a)/\sqrt{w}) + \Phi((-h-a)/\sqrt{w}).$$

The even function $B_{a,w}$ increases with $|h|$. If $q_H = m^{-1}\#\{i : |H_T(i)| \le b_0\}$ and $c_b = -\log B_{a,w}(b_0)$, then

$$\mathbb{P}(q_H \ge q \mid A) \le \min\{1, e^{D - mc_b q}\}, \qquad \mathbb{E}[q_H \mid A] \le \frac{D+1}{mc_b}.$$

Now $1 - B_{a,w}(b_0) = \overline{\Phi}(t) - \overline{\Phi}(t + 2a/\sqrt{w}) \ge \overline{\Phi}(t)/2$. Indeed, for $t, c \ge 0$, changing variables in the Gaussian integral gives

$$\overline{\Phi}(t+c) = \int_t^\infty \phi(s+c)\, ds \le e^{-ct - c^2/2}\overline{\Phi}(t).$$

Apply this with $c = 2a/\sqrt{w}$ and the hypothesis $e^{-2a^2/w} \le 1/2$. Integrating the normal density over $[t, t + 1/(t+1)]$ gives $\overline{\Phi}(t) \ge c_0 e^{-t^2/2}/(t+1)$. Since $-\log B \ge 1 - B$, these estimates give the first term on the right of (33) as an upper bound for $\mathbb{E}[q_H \mid A]$.

The enlarged posterior has relative entropy exactly $-\log \mathbb{P}(A) \le D$ from its prior. Data processing gives the same upper bound for the relative entropy of its ξ marginal from $N(0, I_m)$. The entropy variational inequality and the Gaussian moment generating function imply

$$\tfrac{1}{4}\mathbb{E}[\|\xi\|^2 \mid A] \le D + \log \mathbb{E}e^{\|\xi\|^2/4} = D + \tfrac{m}{2}\log 2,$$

so $\mathbb{E}[\|\xi\|^2 \mid A] \le 4D + 2m$. If $|G_T(i)| \le b_0/2$ and $|H_T(i)| > b_0$, then $\sqrt{w}|\xi_i| > b_0/2$. The averaged Markov inequality completes the proof. □

Lemma 7.5 (Posterior output amplitude)**.** *There are constants $b, C > 0$, depending only on L, a, such that every event A implying $|G(x)| > a$ for all inputs and satisfying $-\log \mathbb{P}(A) \le e^{.045d}$ obeys*

$$\mathbb{E}\Big[\mathbb{P}_x\{|G(x)| \le bd^{(1-\beta)/2}\} \,\Big|\, A\Big] \le C/d \tag{34}$$

for sufficiently large d.

Proof. Increase $-\log \mathbb{P}(A)$ to a deterministic upper bound $D_0 \ge 1$ with $D_0 \le e^{.045d}$. Independently draw an auxiliary uniform sample T of size $m = \lfloor e^{.05d} \rfloor$. Let u be odd with $u/d \to 1/50$, and put $w = \tau_u/2 \asymp d^{-\beta}$. Lemma 5.2 and the spectral estimates imply, with exponentially high probability over T,

$$Q_{u,TT} \succeq (\tau_u/2) I_m, \qquad K_T \succeq w I_m.$$

This event excludes repetitions. Indeed its error terms decay exponentially because $.05 < h(1/50)$ and $.10 < \log 2 - h(1/4)$; the far-pair error also decays exponentially since u is a fixed positive fraction of d.

Lemma 7.4 applies for every such T, to the same full event A. Here $\log(m/D_0) \ge .004d$ for all large d, so $b_0^2 \ge cd^{1-\beta}$. Its first error term is exponentially small; the other terms are $O(1/d)$. Fix $b > 0$ small enough that $bd^{(1-\beta)/2} \le b_0/2$.

Although the enlarged Gaussian splitting depends on T, its marginal law of the full G is always the original Gaussian law, and its conditional marginal is always the same actual posterior $G \mid A$. Averaging the lemma over the independent auxiliary sample therefore changes its left side exactly into the left side of (34). Bad auxiliary samples contribute at most their exponentially small probability. This proves the claim. □

On the sample event in Lemma 7.1, both its adaptive and rough bounds are at most $e^{.045d}$ for large d when $n \le e^{.04d}$. For the adaptive bound this uses $D_r \le \exp[(h(1/200) + o(1))d]$ and $h(1/200) < .04$. Consequently Lemma 7.5 applies to the exact margin-and-label posterior, uniformly over all labelings on that event.

8 Positive subcritical exponents

We now prove the lower bound for $0 < \gamma < 1$. The argument applies simultaneously to every energy-minimizing sign function, so no rule for choosing among ties is required.

Theorem 8.1 (All minimum-energy signs have large error)**.** *For fixed $L \ge 2$ and $a > 0$, there exist $c > 0$ and d_0 such that, for every $d \ge d_0$ and every specified $0 \le n \le e^{cd}$, with probability at least $3/4$ over the training inputs, every minimizer of*

$$\min\left\{\frac{1}{2}\|z\|_K^2 : |z(x)| \ge a \text{ for every } x, \quad f(X_i)z(X_i) \ge a \text{ for every } i\right\}$$

has parity error at least $3/8$. On this event, for every fixed $0 < \gamma < 1$, the actual network posterior satisfies $\liminf_{N\to\infty} F_N(S) \ge 1/16$.

Proof. Apply Lemma 7.2 with its fixed cutoff $r = 1$ and $A = a$. Since $\tau_1 \asymp 1$ and $\widetilde{K} \preceq K$, its interpolant has K-norm squared at most $C_{L,a}(d + nd^{\beta})$. By optimality this gives, except on a set of sample probability $e^{-\Omega_L(d)}$, the simultaneous bound

$$B := \|z\|_K^2 \le C_{L,a}(d + nd^{\beta}) \tag{35}$$

for all minimizers. The statement includes $n = 0$, using the constant comparator. If $n \le \exp[(\log(d+1))^3]$, Lemma 6.3 therefore gives error $1/2 - o(1)$, which is at least $3/8$ for large d.

For the remaining sample sizes, fix $\delta = 1/20$, choose A at least as large as the constant required by Lemma 4.4, and use the cutoffs and notation of Lemma 6.4. Work on the intersection of the comparator event, the squared-coercivity event (19), and the tail-Gram event (29). Its probability is at least $1 - 2\delta - e^{-\Omega_L(d)} > 3/4$ for large d. The range of n absorbs the additive d in (35), giving $B \le C_{L,a} n d^\beta$.

For an arbitrary minimizer write the orthogonal Fourier decomposition

$$z = P + b + t, \qquad P = z_{\le r}, \quad b = z_{(r,u]}, \quad t = z_{>u}.$$

At each observed input where $fP < 0$, training consistency implies $(fP)_- \le |z - P|$. The squared sampling bound, the RKHS evaluation bound, and the tail-Gram event therefore give

$$\frac{1}{4}\|P\|_2^2 \le \frac{1}{n}\sum_i (f(X_i)P(X_i))_-^2 \le \frac{1}{n}\sum_i (z(X_i) - P(X_i))^2 \le \frac{B}{n} b_r \tau_r.$$

Consequently,

$$\|P\|_2^2 \le C_{L,a}\rho^{-\beta} b_r, \tag{36}$$

$$\|b\|_2^2 \le B\ell_r \le C_{L,a} A \rho^{-\beta} H_r, \tag{37}$$

$$\|t\|_2^2 \le B\ell_u \le C_{L,a} d^\beta d^{-10}\tau_r = o(1). \tag{38}$$

All bounds are uniform over the minimizers and the specified sample range. The middle-band norm need not vanish; it is included in the polynomial to which the clipped-parity estimate is applied.

Put $h = P + b$. Its degree is at most u, and orthogonality gives

$$\left(\frac{u}{d}\right)^2 \|h\|_2^2 \le C_{L,a,\delta}\left[\rho^{2-\beta} + \sqrt{A}\,\rho^{3/2-\beta}\sqrt{H_r} + A\rho^{2-\beta} H_r\right] + o(1). \tag{39}$$

In particular the $\rho^{-1/2}$ loss in the tail-Gram bound has been retained. Because $0 < \beta \le 1/2$, every power of ρ on the right is positive. Choose the upper cutoff fraction α sufficiently small, after fixing these constants, that $u\|h\|_2/(ad) \le 1/8$ for all sufficiently large d. Decrease c in Lemma 6.4 accordingly. By (38), also $\|t\|_2/a \le 1/8$. Since $|z| \ge a$ everywhere, Lemma 6.1 now yields

$$|\mathbb{E} f \operatorname{sign} z| \le \frac{1}{4}, \qquad \mathbb{P}_x\{\operatorname{sign} z(x) \ne f(x)\} \ge \frac{3}{8}.$$

This proves the assertion about every minimizer.

For any probability mixture of these minimizing signs, let $q(x)$ be the probability of the correct label and $R = \mathbb{E}_x(1 - q(x))$. The result just proved gives $R \ge 3/8$, independently of the mixture weights. With $F = \mathbb{P}_x\{q(x) \le 2/3\}$, one always has

$$R \le \frac{1-F}{3} + F, \qquad F \ge \frac{3R-1}{2} \ge \frac{1}{16}.$$

Finally, Proposition 2.1 shows that for each fixed $0 < \gamma < 1$, the actual finite-width posterior puts mass $1 - o(1)$ on these signs. Its draw error is therefore at least $3/8 - o(1)$, without requiring convergence of the relative weights among ties. Apply the same elementary inequality at every N and take the lower limit to obtain $\liminf_{N\to\infty} F_N(S) \ge 1/16$. □

9 The fixed-margin Gaussian endpoint

We now prove the confidence lower bound for the exact $\gamma = 0$ posterior. The full event A_S from Section 7 uses the parity labels $y_i = f(X_i)$. Write $\mathbb{E}_S^*$ for expectation under $G \mid A_S$, and let $\mathcal{D} = \mathcal{D}(S)$. Its relative entropy from $N(0, K)$ is exactly $\mathcal{D}$.

Lemma 9.1 (Gaussian quadratic entropy)**.** *Suppose Z is a linear image of a centered Gaussian vector under a reference law, with covariance $T \succeq 0$, trace t, and largest eigenvalue $\ell > 0$. Under any law of relative entropy at most D_0 from that reference law,*

$$\mathbb{E}\|Z\|^2 \le 2t + 4\ell D_0.$$

Proof. The entropy variational inequality states that $\mathbb{E}_Q V \le \mathrm{KL}(Q\|P) + \log \mathbb{E}_P e^V$. It follows immediately from nonnegativity of the relative entropy from $e^V dP/\mathbb{E}_P e^V$, first for bounded V and then by truncation. At $V = \|Z\|^2/(4\ell)$, the Gaussian moment generating function gives

$$\log \mathbb{E}_P e^V = -\frac{1}{2}\sum_j \log\left(1 - \frac{\lambda_j(T)}{2\ell}\right) \le \frac{t}{2\ell},$$

using $-\log(1-x) \le 2x$ for $0 \le x \le 1/2$. The claim follows. The zero-covariance case is immediate. □

For any odd cutoff r, apply this lemma to the high-degree projection $G_{>r}$ as a vector scaled by $M^{-1/2}$, and to its restrictions at the n training locations scaled by $n^{-1/2}$. The resulting exact bounds are

$$\mathbb{E}_S^*\|G_{>r}\|_2^2 \le 2\tau_r + 4\lambda_{r+1}\mathcal{D}, \tag{40}$$

$$\mathbb{E}_S^*\mathbb{P}_n G_{>r}^2 \le 2\tau_r + \frac{4\|Q_{r,SS}\|\mathcal{D}}{n}. \tag{41}$$

Here $\|g\|_2^2 = \mathbb{E}_x g(x)^2$ and $\mathbb{P}_n g = n^{-1}\sum_i g(X_i)$. The normalized empirical covariance has trace exactly τ_r, even if locations repeat. Neither formula asserts independence under the posterior, or log-concavity of the globally conditioned law.

Theorem 9.2 (Endpoint confidence lower bound)**.** *For fixed $L \ge 2$ and $a > 0$, there are $c > 0$ and d_0 such that, for every $d \ge d_0$ and each integer $0 \le n \le e^{cd}$, with probability at least $3/4$ over independent uniform parity training inputs,*

$$R_S := \mathbb{E}_S^*\mathbb{P}_x\{f(x)G(x) < 0\} \ge \frac{3}{8}.$$

Consequently the fraction of inputs whose correct-label posterior probability is at most $2/3$ is at least $1/16$.

Proof. We first consider $1 \le n \le \exp[(\log(d+1))^3]$. On the exponentially high-probability event of Lemma 7.1, $\mathcal{D} \le C(d^2 + nd^{1+\beta})$. Let s be odd with $s \sim d^{1/3}$ and put $p = G_{\le s}$. The spectral estimates give $\tau_s = o(1)$ and

$$\mathcal{D}\lambda_{s+1} \le \frac{\mathrm{poly}(d)\exp[(\log(d+1))^3]}{\binom{d+1}{s+2}} = o(1),$$

uniformly in this range of sample sizes. Thus (40) implies $\mathbb{E}_S^*\|G - p\|_2^2 = o(1)$. Since $|G(x)| > a$ at every input, the signs of G and p can disagree only where $|G - p| \ge a$, using any fixed sign

convention at $p = 0$. Their posterior-averaged disagreement fraction is therefore $o(1)$. Lemma 6.2 gives parity error $1/2 - o(1)$ for every degree-s polynomial threshold, so $R_S \geq 1/2 - o(1) > 3/8$.

For the remaining range, fix $\delta = 1/20$. Choose a fixed sufficiently large sampling constant A and use the maximal odd cutoff of Lemma 6.4:

$$AD_r \log(2D_r) \leq n < AD_{r+2} \log(2D_{r+2}), \qquad \tfrac{1}{4}(\log d)^2 \leq r \leq \alpha d.$$

Here $\alpha < 1/256$ is a small positive constant to be chosen below; the exponential constant c is then chosen small enough to enforce this upper bound. Write

$$\rho = r/d, \qquad H_\rho = \log(e/\rho), \qquad \tau = \tau_r, \quad \ell = \lambda_{r+1}, \qquad B_r = C_\delta(1 + \sqrt{AH_\rho/\rho} + AH_\rho).$$

Using r/d instead of $r/(d+1)$ in the adaptive lemma changes only fixed constants. With probability at least $1 - 2\delta - o(1)$, the denominator, one-sided squared-loss sampling, and tail Gram events all hold:

$$\mathcal{D}/n \leq Cd\rho^{-\beta}, \tag{42}$$

$$\mathbb{P}_n(fP)_-^2 \geq \tfrac{1}{4}\|P\|_2^2 \quad \text{for every degree-}r\text{ polynomial } P, \tag{43}$$

$$\|Q_{r,SS}\| \leq B_r\tau. \tag{44}$$

For the first inequality use (30), $n \geq AD_r \log(2D_r)$, and $\tau \asymp r^{-\beta}$. Indeed $d^2/n = o(1)$ here and $D_r \log(r+1)/n \leq C$, while the main term is $d^{1+\beta}\tau \leq Cd\rho^{-\beta}$. The other two inequalities follow from (19) and Lemma 6.4, respectively. Fix a sample on these events for the rest of the argument.

Take $u = r + 2\lceil 10 \log(d+1) \rceil$ and decompose the actual posterior output into orthogonal Walsh bands,

$$G = p + b + t, \qquad p = G_{\leq r}, \quad b = G_{(r,u]}, \quad t = G_{>u}.$$

Every posterior draw has $f(X_i)G(X_i) > 0$. Therefore, pointwise at the training inputs, $(fp)_- \leq |G_{>r}|$. Equations (43)–(44) and (41) give

$$\mathbb{E}_S^*\|p\|_2^2 \leq 8\tau + 16B_r\tau\mathcal{D}/n. \tag{45}$$

The middle band has prior trace at most τ and largest normalized eigenvalue at most ℓ. Lemma 9.1 gives

$$\mathbb{E}_S^*\|b\|_2^2 \leq 2\tau + 4\ell\mathcal{D}.$$

Orthogonality, together with the shell estimates $n\ell/\tau \leq CAH_\rho$ and $n\lambda_{u+1}/\tau \leq d^{-10}$ from Lemma 6.4, now implies

$$\mathbb{E}_S^*\|p + b\|_2^2 \leq 10\tau + C\frac{\mathcal{D}\tau}{n}(B_r + AH_\rho), \tag{46}$$

$$\mathbb{E}_S^*\|t\|_2^2 \leq 2\tau_u + 4d^{-10}\frac{\mathcal{D}\tau}{n}. \tag{47}$$

No additional empirical sampling event at u is needed.

Set $A_d = b_0 d^{(1-\beta)/2}$, where $b_0 > 0$ is the fixed constant called b in Lemma 7.5. That lemma applies because $\mathcal{D} \leq e^{.045d}$ on the denominator event. Its conclusion and (47) yield, uniformly over the sample sizes in question,

$$\mathbb{E}_S^*\mathbb{P}_x\{|G(x)| \leq A_d\} \leq C/d, \tag{48}$$

$$\mathbb{E}_S^*\|t\|_2^2/A_d^2 \leq C\{(d\rho^\beta)^{-1} + d^{-10}\rho^{-2\beta}\} = o(1). \tag{49}$$

The last assertion uses $\rho \geq (\log d)^2/(4d)$ and $\beta \leq 1/2$.

Let $\mathrm{clip}(z) = \max(-1, \min(1, z))$. The exact clipped-parity identity in Lemma 6.1, applied to the degree-u polynomial $p + b$, gives

$$|\mathbb{E}_x f(x)\,\mathrm{clip}((p+b)(x)/A_d)| \leq \frac{u}{dA_d}\|p+b\|_2.$$

Clipping is 1-Lipschitz, and $|\operatorname{sign} G - \mathrm{clip}(G/A_d)| \leq \mathbf{1}_{\{|G|\leq A_d\}}$. Consequently Cauchy–Schwarz and the preceding bounds show

$$|\mathbb{E}_S^*\mathbb{E}_x f(x)\operatorname{sign} G(x)| \leq \frac{u}{dA_d}\sqrt{\mathbb{E}_S^*\|p+b\|_2^2} + \frac{\sqrt{\mathbb{E}_S^*\|t\|_2^2}}{A_d} + C/d. \tag{50}$$

The second and third terms tend to zero uniformly. Since $u/r = 1 + o(1)$, (42) and (46) bound the square of the first term by

$$C\rho^{2-2\beta}(1 + \rho^{-1/2}\sqrt{H_\rho} + H_\rho) + o(1). \tag{51}$$

All constants here are independent of d, n, ρ; they may depend on the fixed L, a, A, δ. The worst power is $\rho^{3/2-2\beta}\sqrt{H_\rho}$, which tends to zero as $\rho \downarrow 0$ for every $\beta \leq 1/2$. Choose α small enough and then d large enough that the right side of (50) is at most $1/4$. Thus

$$R_S = \frac{1 - \mathbb{E}_S^*\mathbb{E}_x f(x)\operatorname{sign} G(x)}{2} \geq \frac{3}{8}.$$

The good sample event has probability at least $1 - 2\delta - o(1) > 3/4$. At $n = 0$, global sign symmetry gives correct-label probability $1/2$ at every input and the same risk conclusion directly.

Finally put $q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid A_S\}$ and $F_S = \mathbb{P}_x\{q_S(x) \leq 2/3\}$. Then

$$R_S = \mathbb{E}_x(1 - q_S(x)) \leq \frac{1 - F_S}{3} + F_S, \qquad F_S \geq \frac{3R_S - 1}{2} \geq \frac{1}{16}.$$

This conclusion concerns the actual posterior probability, without identifying it with the sign of a posterior mean. □

10 The Gaussian sign posterior: negative exponents

Let $G \sim \mathcal{N}(0, K)$ on the finite cube, and condition on

$$\mathcal{A}_S = \{f(X_i)G(X_i) > 0 : \ 1 \leq i \leq n\}.$$

Write $m = \mathbb{E}[G \mid \mathcal{A}_S]$ and $q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid \mathcal{A}_S\}$. We use the buffered threshold

$$\theta = \frac{13}{20}, \qquad c_\theta = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right) > 0, \qquad F_\theta(S) = \mathbb{P}_x\{q_S(x) \leq \theta\}.$$

The buffer $\theta < 2/3$ will preserve the strict confidence convention when taking the finite-width limit.

Lemma 10.1 (Gaussian mean and confidence)**.** *If s is the number of distinct training inputs, then*

$$\|m\|_K^2 \leq s \leq n, \qquad f(X_i)m(X_i) > 0.$$

Let $\sigma_x^2 = \mathrm{Var}(G(x) \mid G_S)$, with S listing the distinct inputs. If $\sigma_x > 0$ and $q_S(x) > \theta$, then $f(x)m(x) > c_\theta\sigma_x$. For every Fourier cutoff v, moreover,

$$\mathbb{P}_x\{\sigma_x^2 < \tau_v/2\} \leq \frac{2n\ell_v}{\tau_v}, \qquad \|m - \Pi_{\leq v}m\|_2^2 \leq n\ell_v. \tag{52}$$

Proof. Whiten the training vector as $G_S = K_{SS}^{1/2}\xi$. The sign constraints form a homogeneous cone in ξ. Since a standard Gaussian's radius is independent of its direction, conditioning leaves $\mathbb{E}\|\xi\|_2^2 = s$. Jensen's inequality and Gaussian regression therefore give

$$m_S^{\mathsf{T}} K_{SS}^{-1} m_S \le s, \qquad m = K_{\mathcal{X},S} K_{SS}^{-1} m_S.$$

The squared RKHS norm of the regression vector is the first quadratic form, and its signed training values are positive.

The conditional scalar variable $Y = f(x)G(x)$ has a log-concave density: it is a marginal of a Gaussian restricted to a convex set, by Prékopa's marginal theorem [5]. Its cdf H is also log-concave. For $\mu = \mathbb{E}Y$, Jensen's inequality and the probability integral transform give

$$\log H(\mu) \ge \mathbb{E}\log H(Y) = -1, \qquad \mathbb{P}\{Y > \mu\} \le 1 - e^{-1}.$$

Conditioning first on G_S expresses the density of Y as a mixture of Gaussian densities of common variance σ_x^2. It is bounded above by $(\sqrt{2\pi}\sigma_x)^{-1}$. If $q_S(x) > \theta$, this first forces $\mu > 0$, and then

$$\theta < q_S(x) \le 1 - e^{-1} + \frac{\mu}{\sqrt{2\pi}\sigma_x}.$$

This proves the confidence margin.

To bound residual variances, write $G = G_{\le v} + G_{>v}$ independently. Conditioning additionally on the entire low-degree field can only reduce a Gaussian conditional variance. The remaining covariance is

$$Q_v - Q_{v,\mathcal{X},S} Q_{v,SS}^{\dagger} Q_{v,S,\mathcal{X}}.$$

The subtracted matrix is positive semidefinite, bounded by Q_v, and has rank at most s. Its normalized trace is at most $n\ell_v$. Markov's inequality proves the first bound in (52); this automatically includes the observed points, where the full residual variance vanishes. The second bound follows from Fourier orthogonality and $\|m\|_K^2 \le n$. □

Theorem 10.2 (An exponential lower bound for the sign posterior)**.** *For every fixed depth $L \ge 2$, there is $c_L > 0$ such that, for every sufficiently large d and each integer $0 \le n \le e^{c_L d}$,*

$$\mathbb{P}_S\{F_{13/20}(S) \ge 3/8\} \ge 3/4.$$

Consequently the same lower bound holds for the lim inf *of the finite-width strict-2/3 confidence-failure fraction, for each fixed $\gamma < 0$. The width limit is taken at fixed d and training sample.*

Proof. We give the cutoff estimates explicitly. Fix the sampling-failure probability at a sufficiently small universal constant and let A be the resulting constant in Lemma 4.4. Choose the largest odd r such that

$$n \ge A D_r \log(2D_r).$$

First suppose $n \ge \exp((\log(d+1))^3)$. Then the adaptive cutoff estimates of Lemma 6.4 give $r \ge (\log d)^2/4$ for large d. By reducing c_L we may ensure $r \le \alpha d$, for a fixed $\alpha < 1/256$ to be chosen below. Write $\rho = r/d$ and $L_\rho = \log(e/\rho)$. Maximality and the sharp paired eigenvalue estimates give

$$\frac{n\ell_r}{\tau_r} \le C_A L_\rho, \qquad \|Q_{r,SS}\|_{\mathrm{op}} \le b_r \tau_r, \qquad b_r = C_{L,A}\left(1 + \sqrt{L_\rho/\rho} + L_\rho\right), \tag{53}$$

with probability at least $1-\delta$, where δ is a sufficiently small fixed constant. Indeed, $n < AD_{r+2}\log(2D_{r+2})$, $D_{r+2} \asymp \binom{d+1}{r+2}$, and $\log D_{r+2} \le Cr\log(ed/r)$. The ratio estimate (10) proves the first bound. For the Gram bound use Lemma 5.2, with

$$\frac{nd\, s_{2,r}}{\tau_r^2} \le C_A L_\rho/\rho, \qquad \frac{n\ell_r}{\tau_r} \le C_A L_\rho.$$

The far-pair and collision errors vanish uniformly in this range when c_L is sufficiently small.

Let $p = \Pi_{\le r} m$. On the squared-coercivity event, training consistency and the RKHS evaluation inequality imply

$$\frac{1}{4}\|p\|_2^2 \le \frac{1}{n}\sum_i (f(X_i)p(X_i))_-^2 \le \frac{1}{n}\sum_i (m(X_i)-p(X_i))^2 \le \frac{\|m\|_K^2}{n}\|Q_{r,SS}\|_{\mathrm{op}} \le b_r\tau_r.$$

Choose an odd $u = r + O(\log d)$, with a sufficiently large fixed implicit constant. Compare the sharp eigenvalue bounds at the two endpoints: $\ell_u/\ell_r \le C_L\binom{d+1}{r+2}/\binom{d+1}{u+2}$, up to a bounded power ratio. The binomial ratio is geometrically small in $u-r$, uniformly for $r \le \alpha d$. The sharp level masses and the bound $r \ge (\log d)^2/4$ therefore give

$$\tau_u = (1-o(1))\tau_r, \qquad n\ell_u = o(\tau_r). \tag{54}$$

These estimates are uniform over the sample-size range under consideration: the removed band has trace $O((\log d)/r)\tau_r$, and the geometric decay over $O(\log d)$ levels absorbs all polynomial factors in (53). Put $h = \Pi_{\le u} m$ and $t = m - h$. Orthogonality yields

$$\|h\|_2^2 \le 4b_r\tau_r + n\ell_r \le C_{L,A}\tau_r(b_r + L_\rho), \qquad \|t\|_2^2 = o(\tau_r). \tag{55}$$

Lemma 10.1 and (54) show that outside an $o(1)$ fraction of inputs, $\sigma_x^2 \ge \tau_r/3$ and $|t(x)| \le c_\theta\sqrt{\tau_r}/(2\sqrt{3})$. Set $a_r = c_\theta\sqrt{\tau_r}/(2\sqrt{3})$. At every remaining input where $q_S(x) > \theta$, we have $f(x)h(x) > a_r$. Consequently

$$\mathbb{E} f\ \mathrm{clip}(h/a_r, -1, 1) \ge 1 - 2F_\theta(S) - o(1). \tag{56}$$

For any polynomial h of degree at most u, Lemma 6.1 gives

$$|\mathbb{E} f\ \mathrm{clip}(h/a_r, -1, 1)| \le \frac{\|\Delta h\|_1}{a_r d} \le \frac{u\|h\|_2}{a_r d}, \qquad \Delta = \sum_{j=1}^d D_j.$$

For clarity, the first inequality follows directly by setting $q = fh$ and letting R average the one-coordinate flips. Then $Rq = -q + (2/d)f\Delta h$. Stationarity, Jensen's inequality and the triangle inequality imply $|\mathbb{E}|q+a_r| - \mathbb{E}|q-a_r|| \le (2/d)\|\Delta h\|_1$. Use $\mathrm{clip}(q/a_r, -1, 1) = (|q+a_r| - |q-a_r|)/(2a_r)$. The second inequality uses the Walsh multiplier bound $\|\Delta h\|_2 \le u\|h\|_2$. Thus (55) bounds the absolute value in (56) by

$$C_{L,A,c_\theta}\rho\sqrt{1 + \sqrt{L_\rho/\rho} + L_\rho} + o(1),$$

which tends to zero as $\rho \downarrow 0$. Choose α small enough that it is at most $1/8$ for $0 < \rho \le \alpha$, and then take d large enough to absorb the vanishing errors. This proves $F_\theta(S) \ge 3/8$. The sampling and Gram events have combined probability at least $3/4$ after the initial fixed choices.

It remains to cover smaller samples, including cases where the cutoff r does not exist. If $n < \exp((\log(d+1))^3)$, choose an odd $v \asymp d^{1/3}$. The sharp spectrum gives $n\ell_v/\tau_v = o(1)$, because $\log\binom{d+1}{v+2} \asymp v\log(d/v)$ dominates $\log(n+1)$. Let $p = \Pi_{\le v} m$. By Lemma 10.1, outside an $o(1)$ fraction of inputs, the residual variance is at least $\tau_v/2$ and $|m-p| < c_\theta\sqrt{\tau_v}/(2\sqrt{2})$. Therefore confidence above θ implies $fp > 0$ there. Lemma 6.2, with $v = o(\sqrt{d})$, gives $F_\theta(S) \ge 1/2 - o(1) \ge 3/8$ for every such training sample. For $n = 0$ the conclusion also follows immediately from $q_S = 1/2$. Finally, the conditional width limit and $\theta < 2/3$ give the asserted finite-width $\liminf$ statement. □

11 Proof of the main theorem

The positive-exponent, zero-exponent, and negative-exponent conclusions proved above each hold with probability at least $3/4$ for every specified $n \le e^{cd}$. Decrease c and increase d_0 to satisfy their finitely many requirements. The width-limit propositions transfer these conclusions to the stated neural-network posterior.

For clarity, when the posterior's mean classification error is R_N, the confidence fraction always satisfies

$$R_N = \mathbb{E}_x(1 - q_{N,S}(x)) \le \frac{1 - F_N}{3} + F_N, \qquad F_N \ge \frac{3R_N - 1}{2}.$$

The nonnegative-exponent arguments give $\liminf R_N \ge 3/8$, hence the required $1/16$. The negative-exponent argument instead directly bounds the fraction with limiting confidence at most $13/20 < 2/3$, so convergence of each of the finitely many posterior label probabilities gives the same conclusion. This proves Theorem 1.1.

A A small polynomial-weight parity network

For completeness, full parity on k bits has a one-hidden-layer tanh realization of width $k+1$, with weights $O(k \log(k+1))$ and output margin greater than $1/2$. Let

$$t(x) = \frac{k + \sum_i x_i}{2} \in \{0, \dots, k\}, \quad y_j = (-1)^{k-j}, \quad A = \log(16(k+1)).$$

The step functions $s_j(t) = \operatorname{sign}(t - j + 1/2)$ satisfy

$$y_t = c + \sum_{j=1}^{k} y_j s_j(t), \qquad c = (y_0 + y_k)/2 \in \{0, 1\}.$$

Replace each step by $h_j = \tanh(A(t - j + 1/2))$. Since its argument has absolute value at least $A/2$, its error is at most $2e^{-A}$. A constant neuron $h_0 = \tanh 1$ supplies the term c using output weight $c/\tanh 1$. The resulting output differs from parity by less than $1/8$. First-layer input weights are $A/2$ and bias weights have magnitude at most $Ak/2$; readout weights have magnitude at most 2.

At depth $L \ge 3$, the next hidden layer sums these features and applies tanh, producing the correct sign with magnitude at least $\tanh(7/8) > 1/2$. Each additional hidden layer can preserve this margin by a single weight 2, since $\tanh(2u) > 1/2$ whenever $u > 1/2$. A final readout weight 1 completes the construction. Unused neurons are set to zero. Thus the same polynomial bounds hold at every fixed depth. For any other fixed margin $a > 0$, multiplying the final readout by $2a$ gives margin greater than a with the same polynomial complexity bounds.

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